

AC: Values & Phase Difference

AC Values:

Instantaneous value:

$$I = I_0 \sin \omega t \text{ or } I = I_0 \cos \omega t, \text{ and EMF } E = E_0 \sin \omega t \text{ or } E_0 \cos \omega t.$$

Peak value:

$$I_0 \text{ and } E_0.$$

Average value:

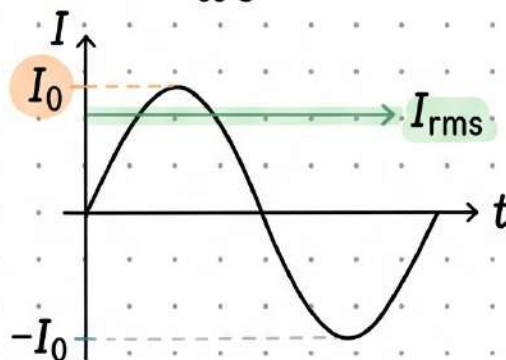
$$I_m \text{ or } I_a = \frac{2I_0}{\pi} = 0.637I_0 \text{ for a half cycle. It is 0 for a full cycle.}$$

$$\text{R.M.S. value: } I_{\text{rms}} = \frac{I_0}{\sqrt{2}} = 0.707I_0.$$

Reactance (X):

$$\text{Inductive reactance: } X_L = \omega L.$$

$$\text{Capacitive reactance: } X_C = \frac{1}{\omega C}.$$



Phase Difference Concepts:

- If the EMF leads the current by $\pi/2$, the reactance is called purely inductive.
- If the EMF lags behind the current by $\pi/2$, the reactance is called purely capacitive.
- If the EMF is in phase with the current, the circuit is called purely resistive.

LCR Impedance & Energy

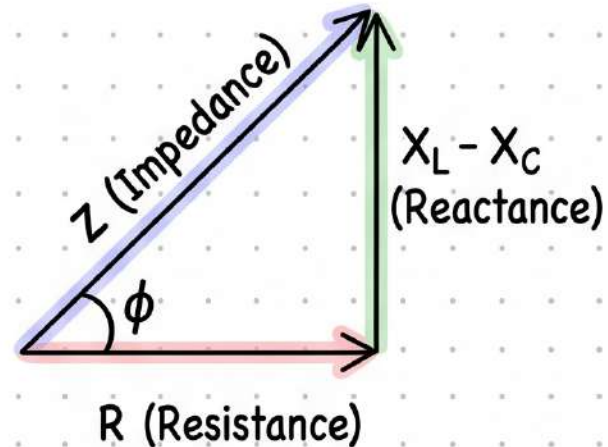
Impedance in LCR Circuit (Z)

Reactance: $X = X_L - X_C = \omega L - \frac{1}{\omega C}$

Impedance formula:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$



Phase angle ($\tan \phi$)

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{IX_L - IX_C}{IR} = \frac{X_L - X_C}{R} = \frac{\omega L - \frac{1}{\omega C}}{R}$$

Sign Convention for LCR Circuit:

- ϕ is positive when $X_L > X_C$.
- ϕ is negative when $X_L < X_C$.
- ϕ is zero when $X_L = X_C$.

Energy Stored

Energy stored in an inductor: $U = \frac{1}{2} LI_0^2$.

Energy stored in a capacitor: $U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{q_0^2}{C} = \frac{1}{2} q_0 V$.

Resonance & Q-Factor

Resonance Conditions:

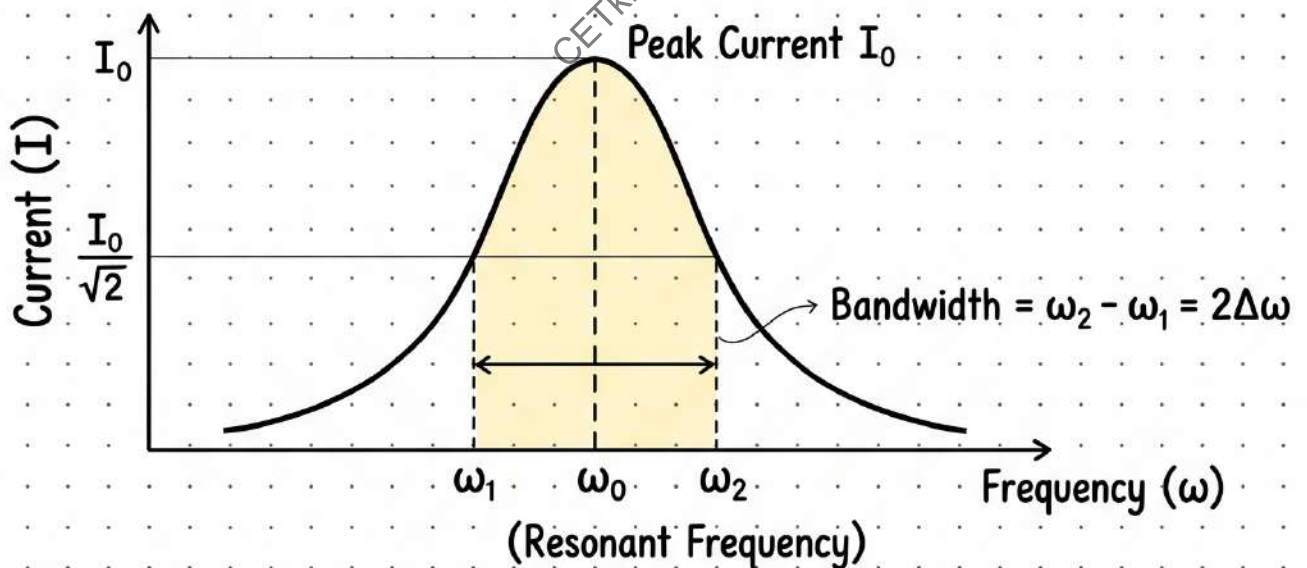
$$X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C}$$

$$\text{Resonant frequency: } \omega = \omega_0 = \frac{1}{\sqrt{LC}}$$

At Series Resonant Frequency:

- Impedance $Z = R$ (minimum value)
- Peak current $I_0 = \frac{E_0}{R}$ (maximum value)
- Phase difference $\phi = 0$ (Current I and Voltage E are in phase)
- Voltage across inductor V_L and capacitor V_C are equal and opposite, $V_L = -V_C$
- Total potential drop across L and C is zero. Applied voltage $E = V_R$

Current vs. Frequency (Resonance Curve)



Bandwidth & Quality Factor:

$$\text{Bandwidth} = \omega_2 - \omega_1 = 2\Delta\omega$$

$$\text{Sharpness of resonance} = \frac{\omega_0}{2\Delta\omega} = \frac{\omega_0 L}{R}$$

$$\text{Q factor: } Q = \frac{\text{Voltage across } L \text{ or } C}{\text{Applied voltage}}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

AC Power & Transformers

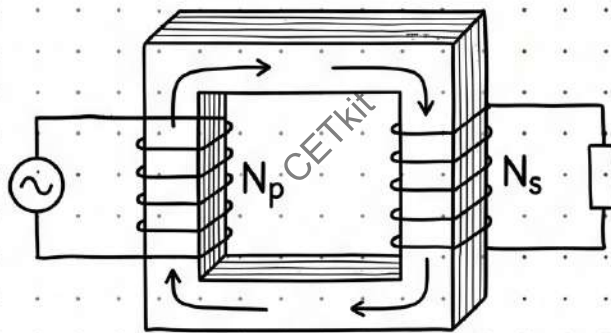
Power in AC Circuit

Power in LCR circuit is $P = EI = E_0 I_0 \sin \omega t \sin(\omega t - \phi)$

- Virtual power component: $\frac{1}{2} E_0 I_0 \cos(2\omega t + \phi)$ → It has frequency twice as that of A.C. Its value over the complete cycle is zero.
- Real power component: $\frac{1}{2} E_0 I_0 \cos \phi$ → It dissipates power and $\cos \phi$ is called the power factor.

Simplified Power equation: $P = E_{\text{rms}} \times I_{\text{rms}} \times \frac{R}{Z}$

Transformers



Transformer ratio (k): $\frac{E_s}{E_p} = \frac{N_s}{N_p} = k$

- For a step-up transformer, $k > 1$ ($N_s > N_p$, $E_s > E_p$).
- For a step-down transformer, $k < 1$ ($N_s < N_p$, $E_s < E_p$).

Efficiency (η) $\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{E_s I_s}{E_p I_p}$

For an ideal transformer, $\eta = 1 = 100\%$, therefore $E_s I_s = E_p I_p$ and $\frac{I_s}{I_p} = \frac{N_p}{N_s} = \frac{1}{k}$

In a step-up transformer, current in the secondary is less than in the primary ($I_s < I_p$), while in a step-down, $I_s > I_p$.